

A
Minor Project Report on
AI Based Electricity Monitoring System

Submitted to



RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL (M.P)

In Partial fulfillment for the award of degree

of

BACHELOR OF TECHNOLOGY

IN

Information Technology

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DECLARATION

We hereby declare that the work, which is being presented in this dissertation entitled “AI based electricity monitoring system” in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in Information Technology, is an authentic record of work carried out by us.

The matter embodied in this report has not been submitted by us for the award of any other degree.

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CERTIFICATE

This is to certify that the project entitled

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PROJECT GUIDE

HEAD OF THE DEPARTMENT

PRINCIPAL



ABSTRACT

This project focuses on developing an innovative system for monitoring and predicting energy consumption. Traditional energy management systems often lack real-time monitoring and personalized recommendations, leading to inefficiencies and wasted resources. In contrast, this project aims to create a smart energy monitoring and prediction system that provides real-time tracking, device identification, and energy forecasting.

The system's objective is to deliver improved energy efficiency, enhanced security, and adaptability. This involves integrating IoT devices, machine learning algorithms, and data visualization tools to create a user-friendly and highly customizable system.

The approach includes defining functional and non-functional requirements, designing the system architecture, and using tools like Python, TensorFlow, and Tableau to manage and analyze data. The outcome will be a lightweight, flexible system demonstrating the potential of smart energy management for targeted environments.



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